

Task 1 DDL

```
CREATE TABLE Department (  
    DNUM NUMBER(9) PRIMARY KEY,  
    DNAME VARCHAR2(50) NOT NULL  
);
```

```
CREATE TABLE Employee (  
    SSN NUMBER(9) PRIMARY KEY,  
    FNAME VARCHAR2(30),  
    LNAME VARCHAR2(30),  
    DOB DATE,  
    GENDER VARCHAR2(10),  
    SUPER_SSN NUMBER(9),  
    DNUM NUMBER(9),  
    CONSTRAINT FK_EMP_SUP FOREIGN KEY (SUPER_SSN) REFERENCES Employee(SSN),  
    CONSTRAINT FK_EMP_DEPT FOREIGN KEY (DNUM) REFERENCES Department(DNUM)  
);
```

```
ALTER TABLE Department ADD (  
    MGR_SSN NUMBER(9),  
    HIRING_DATE DATE,  
    CONSTRAINT FK_DEPT_MGR FOREIGN KEY (MGR_SSN) REFERENCES Employee(SSN)  
);
```

```
CREATE TABLE Dept_Loc (  
    DNUM NUMBER(9),  
    LOCATION VARCHAR2(50),  
    CONSTRAINT PK_DEPT_LOC PRIMARY KEY (DNUM, LOCATION),
```

```
CONSTRAINT FK_DEPT_LOC FOREIGN KEY (DNUM) REFERENCES Department(DNUM)
);
```

```
CREATE TABLE Project (
    PNUM NUMBER(9) PRIMARY KEY,
    PNAME VARCHAR2(50),
    LOCATION VARCHAR2(50),
    CITY VARCHAR2(50),
    DNUM NUMBER(9),
    CONSTRAINT FK_PROJ_DEPT FOREIGN KEY (DNUM) REFERENCES Department(DNUM)
);
```

```
CREATE TABLE Dependent (
    DEPNAME VARCHAR2(50),
    SSN NUMBER(9),
    DOB DATE,
    GENDER VARCHAR2(10),
    CONSTRAINT PK_DEPENDENT PRIMARY KEY (SSN, DEPNAME),
    CONSTRAINT FK_DEP_EMP FOREIGN KEY (SSN) REFERENCES Employee(SSN)
);
```

```
CREATE TABLE Working_Hours (
    PNUM NUMBER(9),
    SSN NUMBER(9),
    HOURS NUMBER(5,2),
    CONSTRAINT PK_WORKING_HOURS PRIMARY KEY (PNUM, SSN),
    CONSTRAINT FK_WH_PROJECT FOREIGN KEY (PNUM) REFERENCES Project(PNUM),
    CONSTRAINT FK_WH_EMPLOYEE FOREIGN KEY (SSN) REFERENCES Employee(SSN)
);
```

